

Shubh Gupta

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SUMMARY

Pre-final year CS undergrad at VIT Bhopal specialising in Machine Learning with published research in real-time computer vision systems. Experienced in building end-to-end ML pipelines with hands-on experience deploying end-to-end ML applications using Streamlit.

EDUCATION

VIT Bhopal University

Bachelor of Technology in Computer Science | CGPA: 8.10

Bhopal, India

Sept. 2023 – May 2027

TECH STACK

Languages: Python, C/C++, JavaScript

ML & Data Science: PyTorch, Scikit-learn, XGBoost, LightGBM, Pandas, NumPy, Matplotlib, Seaborn

Tools & Platforms: Git, Linux/WSL, Jupyter Notebook, MySQL, Streamlit

Web: HTML5, CSS3

PROJECTS

Customer Churn Prediction | Python, LightGBM, Scikit-learn | GitHub | Live App

- Developed an end-to-end ML pipeline to identify at-risk telecom customers, performing extensive EDA to uncover key churn drivers including contract type and tenure.
- Resolved significant class imbalance using SMOTE and built a ColumnTransformer pipeline for streamlined preprocessing.
- Optimised LightGBM classifier using GridSearchCV, achieving a **ROC-AUC score of 0.83**.
- Deployed as an interactive web application using Streamlit, accessible live on Streamlit Cloud.
- Tech: Python, LightGBM, Scikit-learn, SMOTE, Pandas, Seaborn, Streamlit

Delhi AQI Prediction Model | Python, XGBoost, Scikit-learn | GitHub | Live App

- Executed a full data science workflow from EDA to feature engineering and model training on Delhi air quality data.
- Resolved critical overfitting using hyperparameter tuning with GridSearchCV on an XGBoost regression model.
- Achieved a superior **R² score of 0.93+**, showcasing expertise in regression modelling and ensemble techniques.
- Deployed as a live Streamlit web app, enabling real-time AQI predictions from user inputs.
- Tech: Python, XGBoost, Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Streamlit

PUBLICATIONS

Real-Time Data Optimisation for Object Detection via Information Theoretic Network Coding | SSRN

- SSRN Electronic Journal – Engineered a smart traffic system using YOLOv8 and Sparse RLNC, reducing intersection waiting times by **26.9%** and packet loss from 12.4% to **3.9%**.

Improving Object Detection Efficiency via Joint Information Theory and Network Coding | SSRN

- SSRN Electronic Journal – Achieved a **9.03%** accuracy improvement and **7.48%** reduction in packet loss under simulated network congestion.

POSITIONS OF RESPONSIBILITY

Vice-Captain & Club Co-ordinator – Sports Club (Football)

April 2024 – Present

VIT Bhopal University

Bhopal, India

- Successfully co-ordinated major college fests including AdvITYa 2024 & 2025 and Aarambh 2024 & 2025, overseeing event planning, logistics, and execution for 1000+ attendees.
- Achieved Runner-up at AdvITYa 2025, 4th place at SPARDHA 2024 (IIT BHU), and represented VIT at SPORTIKON 2025 (Bennett University).